

Digital Systems & Microcontrollers

Quiz-2 Answer Key

Question-1

Design a synchronous sequential circuit using T Flip-flops that detects the sequence 101 in a serial input stream. The output y , should be 1 only at the clock edge following the third bit of the detected sequence. The circuit must be self-correcting (i.e., overlapping sequences should be allowed).

Part-a: Draw the state diagram for the sequence detector.

[3 marks]

Solution:

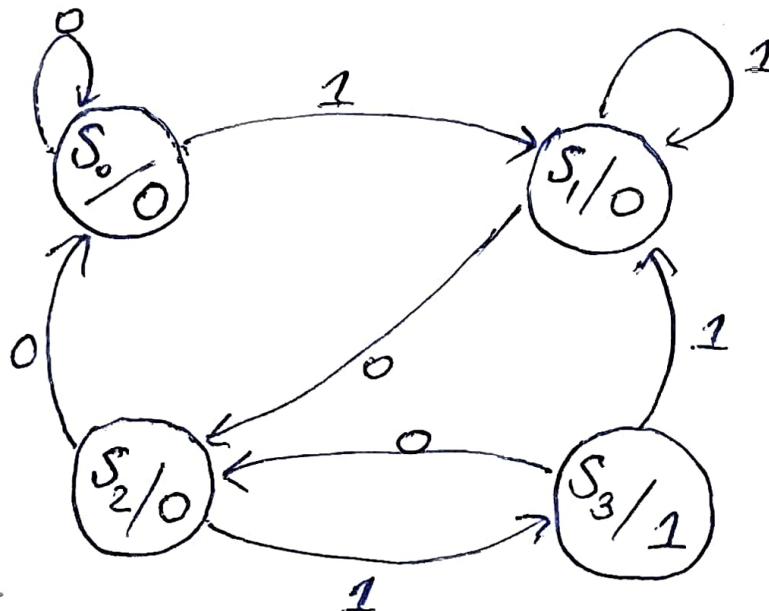


Figure 1: State Diagram

The question explicitly specifies that "the output y should be 1 only at the clock

edge following the third bit of the detected sequence.” This requirement mandates a Moore machine implementation, where the output depends solely on the present state and changes synchronously with the clock edge.

A Mealy machine implementation is **not correct** for this problem because in Mealy machines, the output depends on both the present state and the current input. This means the output can change asynchronously (immediately when the input changes), without waiting for the clock edge. This violates the problem’s explicit requirement of synchronous output change at the clock edge.

Part-b: Create the state table and derive the simplified Next-State equations and the Output equation using Karnaugh Maps (K-maps) or Boolean algebra. [5 marks]

Solution:

State Encoding

- $S_0 = 00$ (Initial state - no sequence detected)
- $S_1 = 01$ (“1” detected)
- $S_2 = 10$ (“10” detected)
- $S_3 = 11$ (“101” detected - output = 1)

Present State		Input X	Next State		Flip-Flop Inputs		Output y
Q_1	Q_0		Q_1^+	Q_0^+	T_1	T_0	
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	0	1	1	0
0	1	1	0	1	0	0	0
1	0	0	0	0	1	0	0
1	0	1	1	1	0	1	0
1	1	0	1	0	0	1	1
1	1	1	0	1	1	0	1

Table 1: State Transition Table for 101 Sequence Detector

K-map for T_1

		Q_0X			
		00	01	11	10
Q_1	0	0	0	0	1
	1	1	0	1	0

From the K-map:

$$T_1 = Q_1Q_0'X' + Q_1'Q_0X' + Q_1Q_0X$$

$$T_1 = (Q_1 \oplus Q_0)X' + Q_1Q_0X$$

K-map for T_0

		Q_0X			
		00	01	11	10
Q_1	0	0	1	0	1
	1	0	1	0	1

From the K-map:

$$T_0 = Q_1'Q_0'X + Q_1'Q_0X' + Q_1Q_0'X + Q_1Q_0X'$$

$$T_0 = Q_0'X + Q_0X' = Q_0 \oplus X$$

K-map for Output y

		Q_0X			
		00	01	11	10
Q_1	0	0	0	0	0
	1	0	0	1	1

From the K-map:

$$y = Q_1Q_0$$

From the K-maps, we derive the following simplified equations:

$$T_1 = (Q_1 \oplus Q_0)X' + Q_1Q_0X \quad (1)$$

$$T_0 = Q_0 \oplus X = Q_0'X + Q_0X' \quad (2)$$

$$y = Q_1Q_0 \quad (3)$$

Part-c: Draw the complete logic circuit diagram.

[2 marks]

Solution:

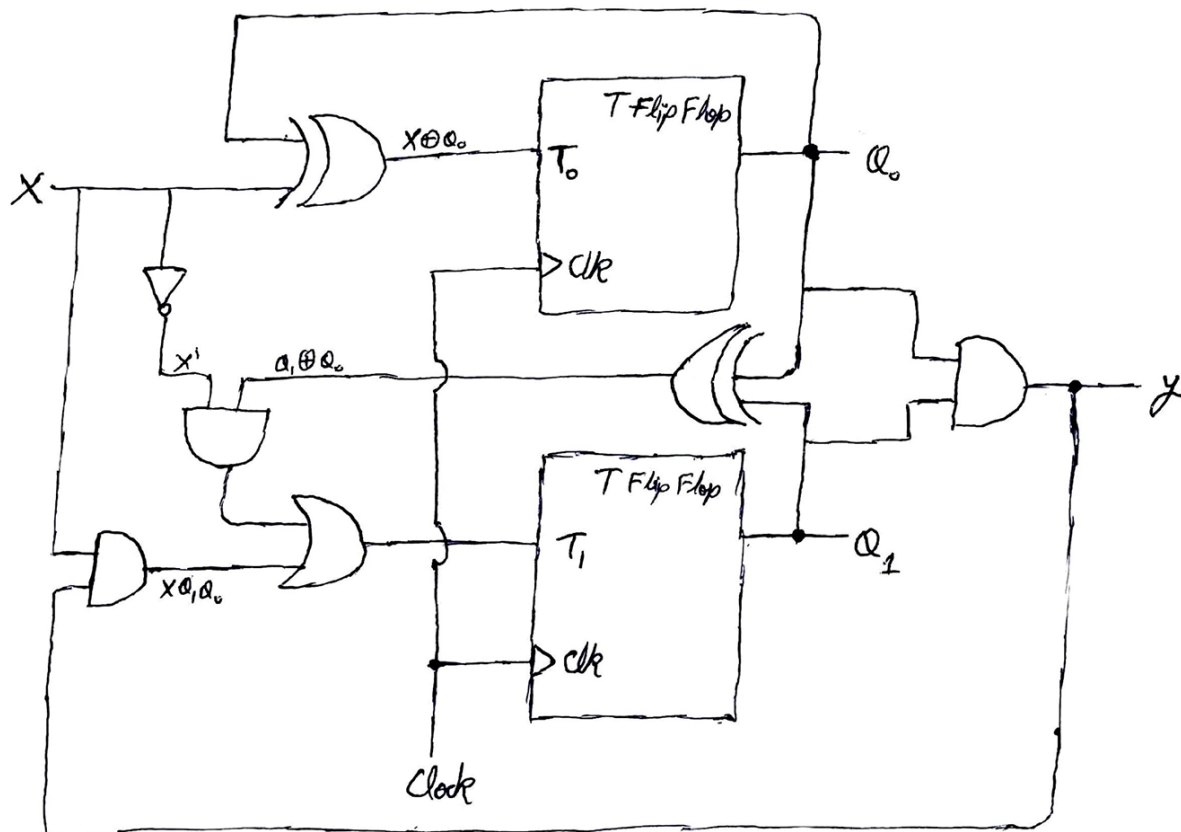


Figure 2: Logic Circuit Diagram

IMPORTANT GRADING NOTE:

- Other methods like synthesis using D or JK flip-flops and then conversion to T flip-flops are also acceptable. But if not converted to T flip-flops, 2 marks will be deducted.
- If Mealy implementation is used, your solution will be graded using the same marking scheme as the Moore machine, following which a 2-mark penalty will be given.

Example: If your Mealy implementation earns 9 marks based on the marking scheme, your final score will be $9 - 2 = 7$ marks.

In other words, Mealy solution will be graded out of **8 marks maximum** instead of 10 marks.

- If "overlapping sequence" is not handled, then 2 marks will be deducted. Basically, such solution will be graded out of a maximum of 8 marks.

Marking Scheme for Question 1:**Part-a: State Diagram [3 Marks]**

- [1 Mark] Correct number and identification of states.
 - Full marks (1): Four states correctly identified (S_0, S_1, S_2, S_3) representing initial, "1" detected, "10" detected, and "101" detected
 - Partial marks (0.5): Incorrect number of states but showing understanding of sequence detection
- [1.5 Marks] Correct state transitions
 - Full marks (1.5): All transitions correctly drawn with proper input conditions ($X=0$ and $X=1$ for each state)
 - Partial marks (1.0): 5-6 transitions correct out of 8 total transitions
 - Partial marks (0.5): 3-4 transitions correct but shows understanding
- [0.5 Marks] Correct output specification
 - Full marks (0.5): Output $y=1$ correctly shown at state S_3 and $y=0$ for other states

Part-b: State Table & Simplified Equations [5 Marks]

- [2.5 Marks] Correct state table construction
 - State encoding (0.5 marks): Correctly assigning binary values to all four states
 - Next state entries (1 mark): All 8 rows of next states correctly determined for both $X=0$ and $X=1$
 - T flip-flop inputs (1 mark): Correctly computing T_1 and T_0 from present and next states using $T = Q \oplus Q^+$
 - Partial credit: Deduct 0.25 marks for each 1-2 errors in state table entries
- [2.5 Marks] Simplified Boolean equations
 - 1 mark for each correct equation T_1 and T_0 , 0.5 marks for y
 - Partial marks: 0.5 marks for correct K-map in case of T_1 and T_0 (if made)

Part-c: Complete Circuit Diagram [2 Marks]

- Full marks (2): Complete and correct logic circuit diagram with all gates, connections accurately represented and labeled input/output.
- Partial marks: 1 mark if part of the circuit correct (like T_1 or T_0), 0.5 deduct for improper labelling or misconnections.

Question-2

Part-a: Draw excitation table for JK FF.

[2 marks]

Solution:

$Q(t)$	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Table 2: Excitation Table for JK Flip-Flop

Part-b: In a JK flip-flop, we have $J=Q'$ and $K=1$. Assuming the flip-flop was initially cleared and then clocked for 6 pulses. Write the sequence at the Q output.[3 marks]

Solution:

JK flip-flop truth table (for reference):

J	K	$Q(t+1)$
0	0	$Q(t)$ (hold)
0	1	0 (reset)
1	0	1 (set)
1	1	$\overline{Q(t)}$ (toggle)

Given: $J = Q'$, $K = 1$, initially $Q(0) = 0$.

Step-by-step table for each pulse:

Pulse n	$Q(t)$ (before)	$J = Q'$	K	$Q(t+1)$
0 (initial)	$Q(0) = 0$	$Q'(0) = 1$	1	$Q(1) = \bar{0} = 1$
1	$Q(1) = 1$	$Q'(1) = 0$	1	$Q(2) = 0$
2	$Q(2) = 0$	$Q'(2) = 1$	1	$Q(3) = 1$
3	$Q(3) = 1$	$Q'(3) = 0$	1	$Q(4) = 0$
4	$Q(4) = 0$	$Q'(4) = 1$	1	$Q(5) = 1$
5	$Q(5) = 1$	$Q'(5) = 0$	1	$Q(6) = 0$

Table 3: Per-pulse calculations showing J , K and the next state.

Final sequence (outputs after each pulse 1–6):

$$Q(1), Q(2), Q(3), Q(4), Q(5), Q(6) = 1, 0, 1, 0, 1, 0.$$

Part-c: What is the need for a reset signal? When will you give a reset for a sequential circuit? **[2 marks]**

Solution: A **reset signal** is used to initialize a sequential circuit to a known starting state before normal operation begins. Since sequential circuits store past information in flip-flops, the initial contents of these storage elements are unpredictable after power-on. Without a reset, the circuit may start in an undefined or unwanted state, leading to incorrect behaviour.

Need for Reset:

- Ensures all flip-flops begin in a known and valid state (usually logic 0).
- Prevents undefined or metastable outputs at startup.
- Provides a controlled reinitialization if the system malfunctions during operation.
- **Conclusion:** In short, a reset signal is needed to initialize a sequential circuit to a known, predictable state.

When to Apply Reset:

- Immediately after power-up to bring the system to a known initial state.
- Whenever the circuit enters an invalid or unused state during operation, or generally if any system fault is observed.
- When a user presses a physical reset button to restart or reboot the system.

Thus, the reset signal is essential for reliable and predictable operation of any sequential circuit.

Part-d: Compare serial and parallel registers.

[2 marks]

Solution: **Registers** are groups of flip-flops used to store and transfer binary data. They can be classified as **serial** or **parallel** based on how data is loaded and read out.

Comparison between Serial and Parallel Registers:

Feature	Serial Register	Parallel Register
Data Loading / Transfer	Data is transferred bit by bit (one bit per clock).	All bits are transferred simultaneously in one clock.
Speed	Slower due to sequential bit transfer.	Faster because all bits move together.
Hardware Complexity	Requires fewer data lines and simpler wiring.	Requires more data lines and complex wiring.
Diagram	Any valid implementation	Any valid implementation

Table 4: Comparison between Serial and Parallel Registers

Marking Scheme for Question 2:**Part-a: Excitation Table [2 Marks]**

- For each correct entry in the table, **0.5 marks** will be awarded.
- Drawing a truth table which covers all possible cases of state transitions will fetch **0.5** marks.

Part-b: Output Sequence [3 Marks]

- For each correct output detected, **0.5 marks** will be awarded.
- **NOTE:** Simply writing the sequence without any sort of explanation as to how the output was obtained will not be given any credit. The answer must be supported by some explanation of how that sequence of outputs was reached.

Part-c: Reset Signal [2 Marks]

- [1 Mark] Need for Reset
 - Mentioning the point in conclusion awards full credit.
- [1 Mark] When to Apply Reset
 - For any correct application, full credit will be awarded.
- **NOTE:** The need for reset and when to apply reset points should be clearly and distinctly mentioned.

Part-d: Comparison of Serial and Parallel Registers [2 Marks]

- [1 Mark] Mentioning the Data Loading/ Transfer point.
- [1 Mark] Mentioning any of the other points listed in the solution, or drawing the diagram.